NAVODIT CHANDRA

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WORK EXPERIENCE

Qualcomm — Hyderabad, India
Senior Machine Learning and Computer Vision Systems Engineer — Dec 2025 – Present
Machine Learning and Computer Vision Systems Engineer — June 2023 – Nov 2025

- Researched and developed computational photography algorithms for **shallow depth-of-field** rendering**, significantly improving bokeh visual quality** for integration into mass-market camera systems
- Designed, trained and optimized a **lightweight** CNN for **real-time** single-image depth estimation**, achieving 30 FPS on-device performance** within **strict power constraints**
- Investigated **mixed-precision quantization** techniques for efficient neural network inference, **reducing latency and power consumption** while maintaining **high prediction accuracy** (**0.0242 EPE**) for deployment on resource-constrained mobile hardware
- Collaborated with **cross-functional teams** to integrate computer vision and deep learning solutions into **camera features**

EDUCATION

Carnegie Mellon University, College of Engineering — Pittsburgh, USA
Master of Science | Specialization in AI and Robotics | GPA: 3.97/4.00 — Dec 2022
Indian Institute of Technology Kanpur — Kanpur, India
B.Tech. in Mechanical Engineering | Minor in Electrical Engineering | GPA: 9.1/10.0 — May 2021

RESEARCH EXPERIENCE

Carnegie Mellon University — Pittsburgh, USA
Graduate Researcher, Mechanical and Artificial Intelligence Lab — May 2022 - Dec 2022

- Refined **image** and **point cloud** feature maps processed by ResNet neural network architecture by introducing **Convolutional Block Attention Module**
- Improved **Driving Score** evaluation metric by **9.5%** by implementing **Additive Attention** for computation of alignment scores in **transformer block** used to combine intermediate image and LiDAR feature maps
- Experimented model performance in simulation by replacing **Self-Attention module** with **Cross-Attention module**

SKILLS

Programming Languages: *Proficient*: Python, C++, *Familiar*: SQL, HTML
Libraries: PyTorch, OpenCV, Gym, NumPy, Pandas, Matplotlib, Scikit-learn
Software and Tools: Linux (Ubuntu), Git, MATLAB, CI/CD

PROJECTS

End to End Learning for Self-Driving Cars — Feb 2022 - Apr 2022

- Developed **CNN** and **CNN-LSTM** models for end-to-end steering angle prediction from sequential image data
- Analyzed trade-offs between **model complexity** and **inference stability**, highlighting implications for **real-time deployment** in video-driven systems

Identification of Abnormal Breasts as Potential Cancers using Machine Learning — Oct 2021 - Dec 2021

- Engineered a **minimal feature set** for tumor classification using **classical ML models**, reducing input dimensionality while preserving predictive performance
- Conducted feature ablation studies to **quantify feature importance and evaluate model robustness**

Seven Segment Digit Recognition using Computer Vision — Mar 2022 - Apr 2022

- Developed an **image processing pipeline** for robust digit extraction from seven-segment displays under varying illumination and noise conditions
- Enhanced **accuracy** by **7.8%** and **speeded up** process of taking readings by **10.4 times** in comparison to average computer typists by utilizing **image processing** operations and **computer vision techniques**

Depth Estimation leveraging Stereo Vision and Generation of 3D Point Cloud — Mar 2022 - Apr 2022

- Implemented **stereo correspondence pipeline** to compute disparity maps and estimate depth from calibrated image pairs

COURSEWORK

Machine Learning and Artificial Intelligence, Deep Learning, Computer Vision, Trustworthy AI Autonomy, Natural Language Processing, Data Structures and Algorithms, Fundamentals of Computing

PATENTS

- US 19/225,642 Content-Aware Image Filtering Operations: **Navodit Chandra**, Gururaj Bhat, Ashish Medewar, Mayukh Roy. Filed on 02-Jun-2025
- US 18/922,132 Kernel Based Blurring: **Navodit Chandra**, Gururaj Bhat, Ashish Medewar. Filed on 21-Oct-2024
- US 18/783,248 Image Processing Using Kernels: **Navodit Chandra**, Gururaj Bhat, Ashish Medewar. Filed on 24-Jul-2024